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$$x_2 = \frac{\Delta x_2}{\Delta} = \frac{-183}{-61} = 3$$

$$x_3 = \frac{\Delta x_3}{\Delta} = \frac{-244}{-61} = 4$$

12.3.4 Application in Economics

Example 12.11

Mr.Anbu, purchased 2 pens, 3 pencils and 1 note book. Mr.Barakath , purchased 4 pens, 3 pencils and 2 notebooks. Mr.Charles purchased 2 pens, 5 pencils and 3 notebooks. They spent ₹32, ₹52 and ₹60 respectively. Find the price of a pen, a pencil and a note book.

Solution:

Let x be the price of a pen, y be the price of a pencil and z be the price of a notebook,

In equations:

$2x + 3y + 1z = 32,$

$4x + 3y + 2z = 52,$

$2x + 5y + 3z = 60$

In matrix form

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 3 & 2 \\ 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 32 \\ 52 \\ 60 \end{bmatrix}$$

$$\begin{aligned} \Delta &= 2(9-10) - 3(12-4) + 1(20-6) \\ &= 2(-1) - 3(8) + 1(14) \\ &= -2 - 24 + 14 = -12 \end{aligned}$$

To find Δ_x

$$\Delta_x = \begin{bmatrix} 32 & 3 & 1 \\ 52 & 3 & 2 \\ 60 & 5 & 3 \end{bmatrix}$$

$$\begin{aligned} \Delta_x &= 32(9-10) - 3(156-120) + 1(260-180) \\ &= 32(-1) - 3(36) + 1(80) \\ &= -32 - 108 + 80 = -60 \end{aligned}$$

To find Δ_y

$$\Delta_y = \begin{bmatrix} 2 & 32 & 1 \\ 4 & 52 & 2 \\ 2 & 60 & 3 \end{bmatrix}$$

$$\begin{aligned} \Delta y &= 2(156-120) - 32(12-4) + 1(240-104) \\ &= 2(36) - 32(8) + 1(136) \\ &= 72 - 256 + 136 = -48 \end{aligned}$$

To find Δ_z

$$\Delta_z = \begin{bmatrix} 2 & 3 & 32 \\ 4 & 3 & 52 \\ 2 & 5 & 60 \end{bmatrix}$$

$$\begin{aligned} \Delta z &= 2(180-260) - 3(240-104) + 32(20-6) \\ &= 2(-80) - 3(136) + 32\ (14) \\ &= -160 - 408 + 448 = -120 \end{aligned}$$

$$x = \frac{-60}{-12} = 5 \quad (\text{Price of a pen})$$

$$y = \frac{-48}{-12} = 4 \quad (\text{Price of a pencil})$$

$$z = \frac{-120}{-12} = 10 \quad (\text{Price of a notebook})$$

Answer checking

$2(5) + 3(4) + 1(10) + 32$

$4(5) + 3(4) + 2(10) + 52$

$2(5) + 5(4) + 3(10) + 60$



Think and Do

Fathima, purchased 6 pens and 5 Pencils spending ₹49, Rani purchased 3 Pens and 4 pencils spending ₹32. What is the price of a pen and pencil?

$$\begin{bmatrix} 6 & 5 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 49 \\ 32 \end{bmatrix}$$

Solution : Price of a pen • ₹4

Price of a pencil • ₹5